

Prathamesh Nikam

Machine Learning / AI Research

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SUMMARY

M.Tech IT graduate with dissertation research in distributed deep learning, ensemble methods, and communication-efficient training. Research projects include hierarchical retrieval and multi-task dialogue modeling, supported by comparative evaluation and a published Python package.

TECHNICAL SKILLS

Research Areas: Distributed Deep Learning, Ensemble Learning, FedAvg, Quantization, NLP, Hierarchical RAG

Methods: Comparative Benchmarks, Model Evaluation, Retrieval Metrics, Weak Supervision, Multi-Task Learning

Tools: Python, PyTorch, Ray, OpenCLIP, FAISS, Scikit-learn, Transformers, NumPy, Pandas, Git

RESEARCH & PROJECTS

HQDE | Hierarchical Quantized Distributed Ensemble [Link]

Python, PyTorch, Ray, ResNet-18, FedAvg

M.Tech dissertation, University of Hyderabad | June 2026 | Supervisor: Prof. P.S.V.S. Sai Prasad

- Developed and published a distributed deep-learning framework with stateful Ray workers, independent ensemble training, and synchronized FedAvg-style updates.
- Benchmarked five image datasets: independent HQDE exceeded centralized accuracy on four, with a largest observed gain of 2.62 percentage points.
- Implemented blockwise delta quantization with error feedback, reducing model-update communication by approximately 3.05x versus matched FP32 FedAvg.

EvidenceMem | Visual Memory and Continual Classification [Link]

Python, PyTorch, OpenCLIP, FAISS, NumPy, MiniBatchKMeans

- Built a compact medoid memory over frozen CLIP embeddings with visual-text fusion, retrieved image evidence, and new-class insertion without encoder updates.
- Reported 89.58% mean CIFAR-10 accuracy across three seeds with 200 stored prototypes, 4.33 percentage points above zero-shot CLIP.
- Evaluated CIFAR-100 classification, OOD detection, memory budgets, and component ablations; documented the accuracy gap to a trained linear probe.

PsychoTA | Multi-Task Dialogue Modeling [Link]

Python, PyTorch, DeBERTa-style Attention, BiGRU

- Implemented a hierarchical dialogue model with speaker-aware disentangled attention, causal cross-utterance attention, and a bidirectional GRU encoder.
- Built heuristic silver-label data preparation and a joint training pipeline for ego state, transaction type, stroke type, and speaker life-position prediction.

EXPERIENCE

University of Hyderabad | Teaching Assistant - Web Development (IT) Lab | Jan - Jun 2026

- Guided students on project design and full-stack application development; recommended GenAI tools for specific project needs, suggested improvements, and supported project evaluation.

University of Hyderabad | Teaching Assistant - Java OOP Lab | Jul - Dec 2025

- Designed lab questions, evaluated student work, and guided students through Java object-oriented programming concepts and debugging.

EDUCATION

University of Hyderabad | M.Tech, Information Technology | 2024 - 2026 | CGPA: 8.68

Modern Education Society College of Engineering | B.E., Computer Engineering | 2019 - 2023

Honors in AI/ML | B.E. CGPA: 8.85

ACHIEVEMENTS

Data Science Conference Hackathon: Winner (2026) | AI4Bharat AWS Hackathon: Finalist | SuperRR Selection Hackathon: Top 36 of 500+ participants (2025) | Finalist in 10+ hackathons held across India.